CHE 201 Fall 2019 HW 8

Paul Diaz

TOTAL POINTS

89.5 / 100

QUESTION 1

1 1 **15 / 15**

✓ **- 0 pts Correct**

- **- 5 pts** no unit
- **- 5 pts** Q incorrect
- **- 2 pts** gallon of hot water incorrect
- **- 15 pts** no submission
- **- 15 pts** Wrong question, problem 4.55

QUESTION 2

2 2 **7.5 / 15**

- **- 0 pts** Correct

✓ **- 7.5 pts Partially wrong or incomplete solution. T2 = 759 °R and T4 = 779 °R**

- **- 15 pts** Wrong solution. T2 = 759 °R and T4 = 779 °R

QUESTION 3

3 3 **17 / 20**

- **- 0 pts** Correct

✓ **- 3 pts T3 unit in K**

- **- 10 pts** Power output
- **- 3 pts** for calculate T3, h should also included h5 and 6
- **- 10 pts** T3 = ?
- **- 5 pts** T3 unfinished
- **- 20 pts** no submission

QUESTION 4

4 4 **15 / 15**

✓ **- 0 pts Correct**

- **- 3 pts** Q is negative. Heat transfer from the system
- **- 2 pts** Final answer should be in MW
- **- 5 pts** Calculation error. Q = -0.21 MW
- **- 5 pts** Delta PE and Delta KE are different than zero
- **- 5 pts** Wrong unit conversion. Q = -0.21 MW
- **- 15 pts** Missing or wrong answer

QUESTION 5

5 5 **15 / 15**

✓ **- 0 pts Correct**

- **- 5 pts** Stage 1: Q = mΔu (closed system energy balance) = 4204.2 kJ
- **- 3 pts** Stage 2: correct equation but wrong solution. Q = +8725.5 kJ. Heat transfer to the system.
- **- 5 pts** Stage 2: Q = Δ(m.u) - heΔm = +8725.5 kJ (energy balance one-exit control volume). Heat transfer to the system.
- **- 15 pts** Missing/wrong answer

QUESTION 6

6 6 **20 / 20**

✓ **- 0 pts Correct**

- **- 1.5 pts** Wrong unit conversion. m1 = 1.2 x 10^5 kg and m2 = 9.3 x 10^5 kg
- **- 5 pts** Wrong energy balance. Should use Eq 4.28
- **- 5 pts** W = 282 GJ
- **- 2.5 pts** Work is positive (done by the compressor)
- **- 20 pts** Wrong answer
- **- 0 pts** Click here to replace this description.

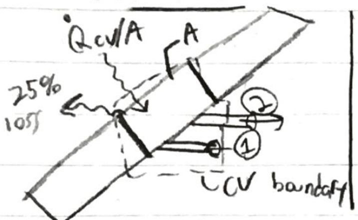
Good job!

CHE 201, HW 8

Paul Diaz

1)

Schematic



Known ← • Incompressible Liquid

- Control volume, Steady state
- $A = 24\,ft^2$, $\dot{Q}_{cv/A} = 200$ Btu/h ft^2 (25% loss)
- 1 inlet/outlet: $T_1 = 90°F$, $P_1 = P_2$, $T_2 = 120°F$
- $\Delta KE = \Delta PE = 0$; 25% heat loss, $\dot{W}_{cv} = 0$

- Find: Gallons of water generated per hour (Volume flow rate)
- $\dot{m}_1 = \dot{m}_2 = \dot{m}$
- Energy balance: $\frac{dE}{dt} = 0 = \dot{Q}_{cv} - \cancel{\dot{W}_{cv}}^{0} + \dot{m}(h_1 - h_2 + \cancel{\frac{V_1^2 - V_2^2}{2}}^{0} + g(\cancel{z_1 - z_2})^{0})$

$$0 = \dot{Q}_{cv} + \dot{m}(h_1 - h_2)$$

$$\dot{Q}_{cv} = 200 \frac{Btu}{h\,\cancel{ft^2}} \cdot 24\cancel{ft^2} \cdot \underbrace{0.75}_{\text{effective energy}} = 3600 \frac{Btu}{hr}$$

$$h_1 \approx h_f(90°F) = 58.07\ BTU/lb$$
$$h_2 \approx h_f(120°F) = 88.00\ BTU/lb$$

$$\dot{m} = -\frac{\dot{Q}_{cv}}{h_1 - h_2} = \frac{-(3600\ BTU/hr)}{-29.93\ BTU/lb} = 120.28\ lb/hr$$

$$\dot{m} = \frac{(A_1 V_1)}{v_1} = \frac{(A_2 V_2)}{v_2} \Rightarrow (A_2 V_2) = \dot{V}_2 = \dot{m} \cdot v_2$$

$$v_2 \approx v_f(120°F) = 0.01621\ ft^3/lb$$

$$\dot{V}_2 = (120.28\ \cancel{lb}/hr) \cdot (0.01621\ \cancel{ft^3}/\cancel{lb}) \cdot \frac{1\ gallon}{0.13368\ \cancel{ft^3}}$$

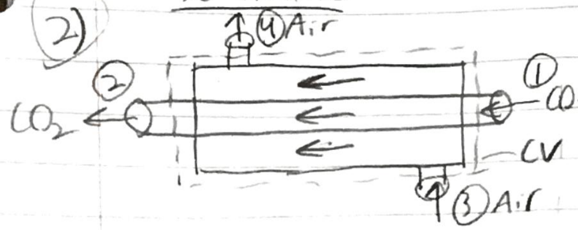
$$\boxed{\dot{V}_2 = 14.59\ gallons/hr}$$

1 1 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 5 pts** no unit
- **- 5 pts** Q incorrect
- **- 2 pts** gallon of hot water incorrect
- **- 15 pts** no submission
- **- 15 pts** Wrong question, problem 4.55

2)

schematic



known

- Both gases are steady state and Ideal Gas
- Parallel heat flow
- $\dot{Q}_{cv} = \Delta PE = \Delta KE = 0$; $\dot{W}_{cv} = 0$

①: $T_1 = 560\,°R$, $\dot{m}_1 = 73.7$ lb/s	② T_2, $P_2 = P_1$
③ $T_3 = 1040\,°R$, $\dot{m}_3 = 50$ lb/s	④ $T_4 = T_2 + 20$, $P_4 = P_3$

- Determine: T_4 and T_2 $(T_4 = T_2 + 20)$
- Mass rates: $\dot{m}_1 = \dot{m}_2 = \dot{m}_{CO_2} = 73.7$; $\dot{m}_3 = \dot{m}_4 = \dot{m}_a = 50$ lb/s
- Energy balance:

$$\frac{dE}{dt} = 0 = \cancel{\dot{Q}_{cv}}^{\,0} - \cancel{\dot{W}_{cv}}^{\,0} + \dot{m}_{CO_2}\left(h_1 - h_2 + \cancel{\frac{V_1^2 - V_2^2}{2}}^{\,0} + g(\cancel{z_1 - z_2})^{\,0}\right) +$$
$$\dot{m}_a\left(h_3 - h_4 + \cancel{\frac{V_3^2 - V_4^2}{2}}^{\,0} + g(\cancel{z_3 - z_4})^{\,0}\right)$$

$$0 = \dot{m}_{CO_2}(h_1 - h_2) + \dot{m}_a(h_3 - h_4)$$
$$\dot{m}_{CO_2}(h_1 - h_2) = \dot{m}_a(h_4 - h_3)$$

- For parallel flow: $h(T_2) = h(T_4) = h$

$$\dot{m}_{CO_2}(h_1 - h) = \dot{m}_a(h - h_3)$$
$$\dot{m}_{CO_2} h_1 - \dot{m}_{CO_2} h = \dot{m}_a h - \dot{m}_a h_3$$
$$-\dot{m}_{CO_2} h - \dot{m}_a h = -\dot{m}_a h_3 - \dot{m}_{CO_2} h_1$$
$$+h(\dot{m}_{CO_2} + \dot{m}_a) = +(\dot{m}_a h_3 + \dot{m}_{CO_2} h_1)$$
$$h = \frac{\dot{m}_a h_3 + \dot{m}_{CO_2} h_1}{\dot{m}_{CO_2} + \dot{m}_a} = \frac{(50\ \text{lb/s})(250.95\ \frac{BTU}{lb}) + 73.7(96.25)}{50 + 73.7}$$

$$h_1 = h_{CO_2}(560\,°R) = 4235.8\ \frac{BTU}{lb\ mol} \cdot \frac{1\ lbmol}{44.01\ lb\ CO_2} = 96.25\ \frac{BTU}{lb}$$
$$h_3 = h_{air}(1040\,°R) = 250.95\ \frac{BTU}{lb}$$
$$h = 158.78\ \frac{BTU}{lb}$$

$h = h_4(T_4) = 158.78 \Rightarrow T_4 =$ Interpolate

Temp	h
660	157.92
T_4	158.78
680	162.73

$$\text{slope} = \frac{162.73 - 157.92}{20\,°R} = \frac{158.78 - 157.92}{T_4 - 660}$$

$$T_4 = \frac{0.86}{4.81} \cdot (20\,°R) + 660\,°R$$
$$T_4 = 663.57\,°R$$

$$T_2 = T_4 - 20 = 643.57\,°R$$

2 2 **7.5 / 15**

- **0 pts** Correct

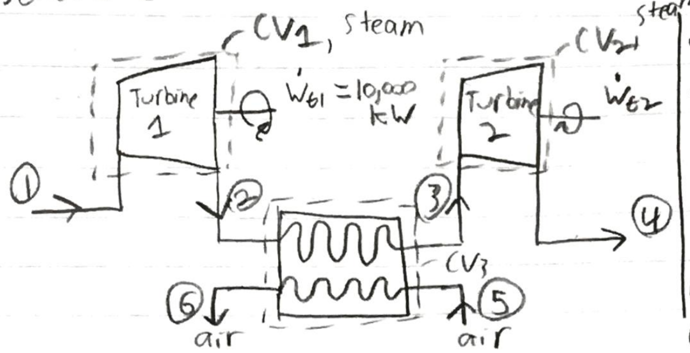
✓ - **7.5 pts** **Partially wrong or incomplete solution. T2 = 759 °R and T4 = 779 °R**

- **15 pts** Wrong solution. T2 = 759 °R and T4 = 779 °R

• Assume air is Ideal Gas

Schematic

3)



Known
- All Control Volume/steady state
- $\dot{Q}_{cv} = \Delta PE = \Delta KE = 0$ for all controle volumes
- ①: $T_1 = 600°C$, $P_1 = 20$ bar
- ② $T_2 = 400°C$, $P_2 = 10$ bar
- ③ $P_3 = 10$ bar
- ④ $t_4 = 240°C$, $P_4 = 1$ bar
- ⑤ $t_5 = 1500k$, $P_5 = 1.35$ bar, $\dot{m}_5 = 1500$ kg/min | ⑥ $T_6 = 1200k$, $P_6 = 1$ bar

• Mass balance: $\dot{m}_1 = \dot{m}_2 = \dot{m}_{t1}$; $\dot{m}_3 = \dot{m}_4 = \dot{m}_{t2}$; $\dot{m}_5 = \dot{m}_6 = \dot{m}_a$

• 3 Energy balance equations

– 1) CV1: $\frac{dE}{dt} = 0 = \cancel{\dot{Q}_{cv}}^0 - \dot{W}_{t1} + \dot{m}_{t1}(h_1 - h_2 + \cancel{\frac{V_1^2 - V_2^2}{2}}^0 + \cancel{g(z_1 - z_2)}^0)$

$$\dot{W}_{t1} = \dot{m}_{t1}(h_1 - h_2)$$

– 2) CV2: $\frac{dE}{dt} = 0 = \cancel{\dot{Q}_{cv}}^0 - \dot{W}_{t2} + \dot{m}_{t2}(h_3 - h_4 + 0 + 0)$

$$\dot{W}_{t2} = \dot{m}_{t2}(h_3 - h_4)$$

– 3) CV3: $\dot{m}_2 = \dot{m}_3 = \dot{m}_s$

$$\frac{dE}{dt} = 0 = \cancel{\dot{Q}_{cv3}}^0 - \cancel{\dot{W}_{cv3}}^0 + \dot{m}_s(h_2 - h_3 + 0 + 0) + \dot{m}_a(h_5 - h_6 + 0 + 0)$$

$$0 = \dot{m}_s(h_2 - h_3) + \dot{m}_a(h_5 - h_6) \Rightarrow \dot{m}_a(h_5 - h_6) = \dot{m}_s(h_3 - h_2)$$

a) Determine T_3 (k)

$$CV1: \dot{m}_{t1} = \frac{\dot{W}_{t1}}{h_1 - h_2}$$

$h_1 = h(600°C, 20\text{ bar}) = 3690.1 \frac{kJ}{kg}$

$h_2 = h(400°C, 10\text{ bar}) = 3263.9 \frac{kJ}{kg}$

$$\dot{m}_s = \dot{m}_2 = \dot{m}_{t1} = \frac{10{,}000 \frac{kJ}{s}}{426.2 \frac{kJ}{kg}} = 23.46 \text{ kg/s}$$

$$CV2: h_3 = \frac{\dot{m}_a}{\dot{m}_s}(h_5 - h_6) + h_2$$

$\dot{m}_a = \dot{m}_5 = 1500 \text{ kg/min} \cdot \frac{1 min}{60 s} = 25 \text{ kg/s}$

$h_5 = h_{air}(1500k) = 1635.97 \frac{kJ}{kg}$

$h_6 = h_{air}(1200k) = 1277.79 \frac{kJ}{kg}$

$$h_3 = \left(\frac{25 \text{ kg/s}}{23.46 \text{ kg/s}}\right)(1635.97 - 1277.79) + 3263.9 \frac{kJ}{kg}$$

$$h_3 = 3645.54 \text{ kJ/kg} = h(10\text{ bar}, T_3)$$

T_3: interpolate between $h(10\text{bar}, 540°C)$ and $h(10\text{bar}, 600°C)$

$(T_3 - 540) \cdot \frac{132.3}{60} = 79.94$

3)

h (10 bar)	Temp
3565.6 $\frac{kJ}{kg}$	540 °C
3645.54	T_3
3697.9	600 °C

$$\text{slope} = \frac{\overbrace{3697.9 - 3565.6}^{132.3\ \frac{kJ}{kg}}}{60\,°C} = \frac{\overbrace{3645.54 - 3565.6}^{79.94\ \frac{kJ}{kg}}}{T_3 - 540}$$

$$T_3 = 79.94\ \cancel{\tfrac{kJ}{kg}} \left(\frac{60\,°C}{132.3\ \cancel{\tfrac{kJ}{kg}}}\right) + 540\,°C$$

$$\boxed{T_3 = 576.25\ °C}$$

b) Find $\dot{W}_{t2}$ (kW)

CV$_2$: $\dot{W}_{t2} = \dot{m}_{t2}(h_3 - h_4)$

$\dot{m}_s = \dot{m}_3 = \dot{m}_{t2} = 23.46$ kg/s

$h_4 = h(240°C,\ 1\ \text{bar}) = 2954.5\ \frac{kJ}{kg}$

$$\dot{W}_{t2} = 23.46\ \cancel{kg}/s\ (3645.54 - 2954.5\ \tfrac{kJ}{\cancel{kg}})$$

$$\boxed{\dot{W}_{t2} = 16212\ kW}$$

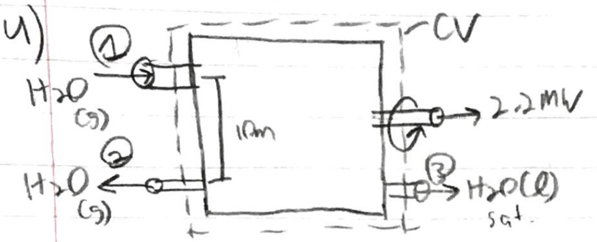
3 3 **17 / 20**

- **- 0 pts** Correct
- ✓ **- 3 pts** **T3 unit in K**
- **- 10 pts** Power output
- **- 3 pts** for calculate T3, h should also included h5 and 6
- **- 10 pts** T3 = ?
- **- 5 pts** T3 unfinished
- **- 20 pts** no submission

Schematic

Knowns

4)



- Control volume, steady state
- 1 inlet / 2 outlets
- Cogeneration system

- Find: $\dot{Q}_{cv}$ (MW)
- mass balance: $\dot{m}_1 = \dot{m}_2 + \dot{m}_3$; $\dot{m}_2 = \dot{m}_3 = \dot{m}_o \Rightarrow \dot{m}_1 = 2\dot{m}_o$

  $\dot{m}_o = 0.75$ kg/s , $\dot{m}_1 = 1.5$ kg/s
- Energy balance:

  $(\sum \dot{m}_i (h + PE + KE) - \sum \dot{m}_o (h + PE + KE))$

$$\frac{dE}{dt} = 0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1\left(h_1 + \frac{V_1^2}{2} + gz_1\right) - \left[\dot{m}_2\left(h_2 + \frac{V_2^2}{2} + gz_2\right) + \dot{m}_3\left(h_3 + \frac{V_3^2}{2} + gz_3\right)\right]$$

$h_1 = h(360°C, 20\text{ bar}) = 3159.3 \frac{kJ}{kg}$

$h_2 = h_g(1\text{ bar}) = 2675.5 \frac{kJ}{kg}$

$h_3 = h_f(1\text{ bar}) = 417.46 \frac{kJ}{kg}$

$\frac{V_1^2}{2} + gz_1 = \frac{(50m/s)^2}{2} + (9.81m/s^2)(10m) = 1348.1 \frac{m^2}{s^2} \cdot \frac{1\,J/kg}{m^2/s^2} \cdot \frac{1\,kJ}{10^3 J}$

$\frac{V_1^2}{2} + gz_1 = 1.3481 \frac{kJ}{kg}$

$\frac{V_2^2}{2} + gz_2 = \frac{(100m/s)^2}{2} + g(0m) = 5000 \frac{m^2}{s^2} \cdot \frac{1\,J/kg}{m^2/s^2} \cdot \frac{1\,kJ}{10^3 J}$

$\frac{V_2^2}{2} + gz_2 = 5$ kJ/kg

$V_3^2/2 + gz_3 = \frac{(100m/s)^2}{2} + g(0m) \cdot \frac{1\,J/kg}{m^2/s^2} \cdot \frac{1\,kJ}{10^3 J} = 5$ kJ/kg

$-\dot{Q}_{cv} = -(2.2MW) + (1.5 kg/s)(3159.3 + 1.3481 \frac{kJ}{kg}) - [0.75 kg/s(2675.5 + 5 kJ/kg) + 0.75 kg/s(417.46 + 5 \frac{kJ}{kg})]$

$-\dot{Q}_{cv} = -2.2MW + 4740.987 kW - (2010.375 + 316.845 kW)$

$-\dot{Q}_{cv} = -2.2 MW + 2413.767 kW \cdot \frac{1 MW}{10^3 kW}$

$-\dot{Q}_{cv} = -2.2 + 2.413767 MW$

$-\dot{Q}_{cv} = 0.213767$ MW

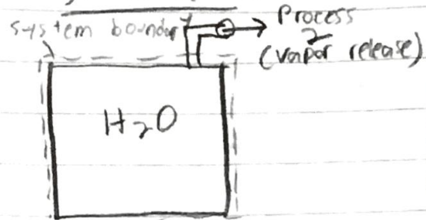
$\therefore$

$\boxed{\dot{Q}_{cv} = -0.213767 MW}$

4 4 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 3 pts** Q is negative. Heat transfer from the system
- **- 2 pts** Final answer should be in MW
- **- 5 pts** Calculation error. Q = -0.21 MW
- **- 5 pts** Delta PE and Delta KE are different than zero
- **- 5 pts** Wrong unit conversion. Q = -0.21 MW
- **- 15 pts** Missing or wrong answer

5) schematic



Known:

- Initial conditions: $V_1 = 0.1\,m^3$, $P_1 = 1$ bar, $x_1 = 0.01$
- Process 1: $V_2 = V_1$, $P_2 = 20$ bar
- Process 3: $\dot{m}_s$ exists, $P_3 = P_2$, only $H_2O(g)$ left
- $\Delta KE = \Delta PE = 0$; control volume

- Find: Heat transfer for each stage (kJ)
- Stage 1: $m_2 = m_1$, $v_1 = v_2$: b/c rigid tank

$$v_1 = \overset{1.0432\cdot 10^{-3}}{v_f(1\,bar)} + \overset{0.01}{x_1}(\overset{1.694}{v_g(1bar)} - v_f(1bar))$$

$$v_1 = 1.0432\cdot 10^{-3}\tfrac{m^3}{kg} + 0.01(1.6929) = 0.01797\tfrac{m^3}{kg} = v_2$$

$$m_{H_2O} = \frac{V}{v_1} = \frac{0.1m^3}{0.01797\frac{m^3}{kg}} = 5.56\text{ kg} \ (m_1 = m_2)$$

- mass rate: $\frac{dm_{cv}}{dt} = 0$ (no loss/gain) ; $\frac{dE}{dt} = \frac{dU}{dt} = \dot{Q}_{cv} - \cancel{\dot{W}_{cv}}^{0} + 0$

Transient analysis: $\Delta U = Q_{cv}$

$$Q_{cv} = \Delta U = m_2u_2 - m_1u_1 = m(u_2 - u_1)$$

$$v_2 = v_1 = 0.01797\tfrac{m^3}{kg} = \overset{1.1767\cdot 10^{-3}}{v_f(20bar)} - \overset{?}{x_2}(\overset{0.09963}{v_g(20bar)} - v_f(20\,bar))$$

$$x_2 = \frac{0.01797 - v_f(20bar)}{v_g(20) - v_f(20\,bar)} = 0.17$$

$$u_1 = \overset{417.36}{u_f(1bar)} + x_1(\overset{2506.1}{u_g(1bar)} - u_f(1bar)) = 438.247\tfrac{kJ}{kg}$$

$$u_2 = \underset{906.44}{u_f(20bar)} + \underset{.17}{x_2}(\underset{2600.3}{u_g(20)} - u_f(20)) = 1194.40\tfrac{kJ}{kg}$$

$$\boxed{Q_1 = (5.56\text{ kg})(1194.40 - 438.247\tfrac{kJ}{kg}) = 4204.19\text{ kJ}}$$

- For Q_2, Unsteady state, Internal NRG changes, mass loss

$$\Delta m = \frac{dm_{cv}}{dt} = -\dot{m}_3 = \Delta m \ ; \ \frac{dU}{dt} = \Delta U$$

Transient analysis

$$\frac{dE}{dt} = \frac{dU}{dt} = \dot{Q}_{cv} - \cancel{\dot{W}_{cv}}^{0} - \dot{m}_3(h_3 + \cancel{\tfrac{V_3^2}{2}}^{0} + \cancel{gz_3}^{0})$$

$$\Delta U = Q_2 + h_3\Delta m$$

$$Q_2 = \Delta U - h_3\Delta m$$

$$m_2 = 5.56\text{ kg} \ ; \ v_3 = v_g(20\text{ bar}) = 0.09963\tfrac{m^3}{kg}$$

$$m_3 = \frac{V_{Tank}}{v_3} = \frac{0.1m^3}{0.09963\frac{m^3}{kg}} = 1.004\text{ kg}$$

$$\Delta m = m_3 - m_2 = 1.004 - 5.56\text{ kg} = -4.556\text{ kg}$$

5) $\Delta U = u_3 m_3 - u_2 m_2$

$u_3 = u_g(20\text{bar}) = 2600.3 \frac{kJ}{kg}$

$u_2 = u_f(20\text{bar}) + x_2(u_g(20\text{bar}) - u_f(20\text{bar}))$

$\hookrightarrow 905.44 \quad \hookrightarrow 0.17 \quad \hookrightarrow 2600.3$

$u_2 = 1194.40 \frac{kJ}{kg}$; $h_3 = h_g(20\text{bar}) = 2799.5 \frac{kJ}{kg}$

$\Delta U = 2600.3(1.004) \frac{kJ}{\cancel{kg}} \cdot \cancel{kg} - (1194.40)(5.56) \frac{kJ}{\cancel{kg}} \cdot \cancel{kg}$

$\Delta U = -4030.1628 \text{ kJ}$

$Q_2 = \Delta U - h_3 \Delta m$

$Q_2 = -4030.1628 \text{ kJ} - (2799.5 \frac{kJ}{\cancel{kg}})(-4.556 \cancel{kg})$

$\boxed{Q_2 = 8724.359 \text{ kJ}}$

5 5 **15 / 15**

✓ **- 0 pts** Correct

- 5 pts Stage 1: Q = mΔu (closed system energy balance) = 4204.2 kJ

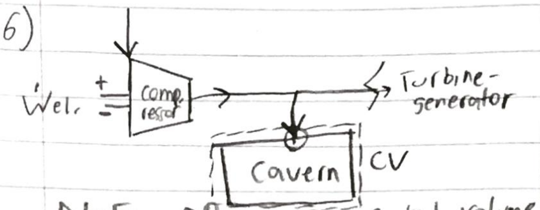
- 3 pts Stage 2: correct equation but wrong solution. Q = +8725.5 kJ. Heat transfer to the system.

- 5 pts Stage 2: Q = Δ(m.u) - heΔm = +8725.5 kJ (energy balance one-exit control volume). Heat transfer to the system.

- 15 pts Missing/wrong answer

6)

Schematic



$\Delta KE = \Delta PE = 0$, Control volume

Known

- Compressed-air energy storage system
- Assume Ideal Gas (Air)
- Cavern data: $V = 10^5\ m^3$

Initial: $T_1 = 290K$, $P_1 = 1\ bar$

Final: $T_2 = 790K$, $P_2 = 21\ bar$

a) determine air mass initially and finally (m_1 and m_2)

- For $m_1 \Rightarrow PV = m\bar{R}T$; $\bar{R} = 8.314\ kJ/kmol\,K \cdot \frac{1\ kmol}{28.97\ kg\ Air}$

$\bar{R} = 0.287\ kJ/kg\,K$

$kPa \cdot m^3 = kJ$

$$m_1 = \frac{P_1 V}{\bar{R} T_1} = \frac{1\,bar\left(\frac{10^2\,kPa}{1\,bar}\right)(10^5\,m^3)}{(0.287\,kJ/kg\,K)(290K)} = 120154.6\ kg$$

$$m_2 = \frac{P_2 V}{\bar{R} T_2} = \frac{21\,bar\left(\frac{10^2\,kPa}{1\,bar}\right)(10^5\,m^3)}{(0.287\,kJ/kg\,K)(790K)} = 926255.24\ kg$$

b) Work required by compressor (GJ)

- Energy balance on cavern ; $\frac{dm_{cv}}{dt} = \dot{m}_i = \Delta m$

$$\frac{dE}{dt} = \frac{dU}{dt} = \cancel{\dot{Q}_{cv}}^{0} - \dot{W}_{cv} + \dot{m}_i\left(h_i + \cancel{\frac{V_i^2}{2}}^{0} + \cancel{gz_i}^{0}\right)$$

$$\frac{dU}{dt} = \Delta U = -W_{cv} + h_i(\Delta m)$$

$$W_{cv} = h_i(m_2 - m_1) - (u_2 m_2 - u_1 m_1)$$

$$\left.\begin{aligned} h_i &= h_1(290K) = 290.16\ \tfrac{kJ}{kg} \\ u_1 &= u(290K) = 206.91\ \tfrac{kJ}{kg} \\ u_2 &= u(790K) = 584.21\ \tfrac{kJ}{kg} \end{aligned}\right\}\ \text{table A-22}$$

$$W_{cv} = 290.16\,\tfrac{kJ}{kg}(926255.24 - 120154.6\,kg) - \begin{bmatrix} 584.21(926255.24) - \\ 206.91(120154.6) \end{bmatrix}$$

$$W_{cv} = 2.339\cdot10^8\,kJ - (5.163\cdot10^8\,kJ)\cdot\frac{1\,GJ}{10^6\,kJ}$$

$$W_{cv} = -282.4\,GJ$$

- The cavern requires 282.4 GJ of energy to transfer pressurize air into it, therefore the compressor needs to do 282.4 GJ of work.

6 6 **20 / 20**

✓ **- 0 pts** Correct

- **- 1.5 pts** Wrong unit conversion. m1 = 1.2 x 10^5 kg and m2 = 9.3 x 10^5 kg
- **- 5 pts** Wrong energy balance. Should use Eq 4.28
- **- 5 pts** W = 282 GJ
- **- 2.5 pts** Work is positive (done by the compressor)
- **- 20 pts** Wrong answer
- **- 0 pts** Click here to replace this description.

Good job!